

The Logic of Partitions and Partition Logics: Ore’s Correspondence, Contextual Pasting, and Direct-Sum Decompositions

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The term “partition logic” denotes two constructions at different levels. In automaton and generalized-urn models, selected partitions generate Boolean event algebras whose contextwise union forms a concrete pasted event structure; in Ellerman’s framework, whole partitions are classifications governed by refinement and partition operations. For a finite set U , Ore’s correspondence maps each generator π to its Boolean algebra $\mathcal{B}(\pi)$, but it neither identifies the pasted carrier with $\Pi(U)$ nor makes pasting a partition operation. It yields $\mathcal{B}(\pi \wedge \sigma) = \mathcal{B}(\pi) \cap \mathcal{B}(\sigma)$ and $\mathcal{B}(\pi \vee \sigma) = \langle \mathcal{B}(\pi) \cup \mathcal{B}(\sigma) \rangle_{\text{BA}}$, where $\langle \cdot \rangle_{\text{BA}}$ denotes Boolean-algebra generation. Thus meet captures the common event algebra, whereas join gives the ambient Boolean closure. Chinese-lantern, Firefly, and triangular examples distinguish shared events, atomic intertwining, and inherited concrete order. Ellerman’s direct-sum decompositions (DSDs) provide a vector-space analogue: component projections of an orthogonal DSD resolve the identity and encode exclusive outcomes, but its components are not equivalence classes of vectors. Gleason and Kochen–Specker applications require globally context-consistent valuations on those projections.

I. INTRODUCTION

The different uses of the expression “partition logic” can be formulated as a type difference. A partition may label an experimental question: its blocks, regarded as subsets of a hidden-state set, support the possible outcome-event propositions, and unions of its blocks support coarser event propositions. Alternatively, the whole partition may itself be the mathematical object, namely a classification determined by which pairs lie in the same block and which are distinguished.

Following established usage, we retain “partition logic” when referring to the automaton and generalized-urn literature. For the construction itself, “contextual partition pasting” or “pasted event structure” is more literal: Boolean event algebras are combined along their common events.

The first use occurs in generalized urn models, finite automata, and contextual event structures [1–4]. A selected family of partitions gives a selected family of Boolean algebras of events; pasting these algebras yields a partial Boolean event structure. Under additional structural conditions it may form an orthoalgebra, an orthomodular poset, or, in special cases, an orthomodular lattice. The second use is Ellerman’s logic of partitions [5–7], which equips the full set of partitions of a universe with operations taking partitions to partitions. Ellerman subsequently linearizes the construction by replacing set partitions with direct-sum decompositions (DSDs) of vector spaces [8].

The two uses have distinct histories, carriers, and operations. For finite U , Ore’s correspondence identifies partitions of U with Boolean subalgebras of $\mathcal{P}(U)$ [9]. Rota and collaborators developed a related logic of commuting

equivalence relations [10]. Ore’s correspondence supplies an exact comparison at the level of context generators on a fixed U : each selected partition π_k corresponds to the Boolean context $\mathcal{B}(\pi_k)$. In the concrete set representation the carrier is then formed as $L = \bigcup_k \mathcal{B}(\pi_k)$. Since this union is generally not a Boolean algebra, it is generally not $\mathcal{B}(\tau)$ for any partition τ . The meet and join of the generating partitions nevertheless summarize, respectively, the common event algebra and the Boolean closure of the chosen concrete context family.

II. THE TWO MATHEMATICAL LEVELS

A. Contextual partition pastings

Let U be a finite set and let

$$\pi = \{B_i\}_{i \in I} \quad (1)$$

be a partition of U . In the contextual interpretation, each block B_i , regarded as a subset of U , supports the unary event proposition $u \in B_i$ associated with one experimental outcome; it is the set of underlying states yielding that outcome. Moreover,

$$\mathcal{B}(\pi) = \left\{ \bigcup_{i \in S} B_i : S \subseteq I \right\} \subseteq \mathcal{P}(U) \quad (2)$$

is the Boolean algebra of its decidable events. For a chosen finite family $\{\pi_1, \dots, \pi_m\}$, the carrier of the associated pasted event structure is

$$L = \bigcup_{k=1}^m \mathcal{B}(\pi_k). \quad (3)$$

Equation (3) defines a concrete set-represented union inside one fixed $\mathcal{P}(U)$: equal subsets are identified and

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TABLE I. The two uses of partitions. Ore’s correspondence applies to the generating partitions and their Boolean context algebras; it does not identify $\Pi(U)$ with the pasted carrier.

Aspect	Whole-partition logic	Contextual partition pastings
Carrier	$\Pi(U)$, all partitions of U	Concrete: $\bigcup_k \mathcal{B}(\pi_k) \subseteq \mathcal{P}(U)$; abstract: amalgamated Boolean blocks
Elements	Whole partitions	Concrete: subset events; abstract: context events modulo prescribed identifications
Order	Refinement, equivalently distinction-set (dit-set) inclusion	Concrete: subset inclusion; abstract: context-generated order
Operations	Join, meet, implication, and closure-based operations	Boolean operations are total locally; ambient operations may leave the concrete carrier, while abstract cross-context operations are partial
Nonclassicality	Nondistributivity of the partition lattice	Partiality and nondistributivity of the pasted event structure
Logical reading	Classification questions or abstract attributes; same/different distinctions	Event propositions; subset membership and contextual incidence

order is inherited from inclusion. This must be distinguished from an abstract pasting of the same Boolean blocks, in which only common elements are identified and order and partial operations are generated contextwise. Ambient inclusion can add further cross-context comparabilities, as the triangle-shaped family in Sec. III C shows. Whether the concrete and abstract constructions agree is example-dependent. Within each context Boolean operations are total. Across contexts, ambient set operations remain defined in $\mathcal{P}(U)$ but may leave L , whereas the corresponding operations of the abstract pasting are partial. Pasting is a construction of an event structure, not a logical connective acting on two partitions.

B. Ellerman’s logic of whole partitions

Ellerman’s carrier is $\Pi(U)$, the set of all partitions of U . On his reading, a partition π represents a classification question or abstract attribute. Its blocks are same-answer or same-value classes, and its distinction set (dit-set) $\text{dit}(\pi)$ consists of the ordered pairs assigned different answers or values. Thus its primitive semantic instance is $(u, u') \in \text{dit}(\pi)$, a pair distinction, rather than $u \in S$, an in/out assertion about one element. Partition variables and formulas range over whole classifications, not over event propositions. We shall therefore distinguish a whole-partition question from the event propositions in $\mathcal{B}(\pi)$ [5, 6].

This is the semantic face of the subset–partition duality: subsets, viewed as subobjects, support unary membership assertions, whereas partitions or equivalence relations, viewed as quotient objects, support binary assertions of sameness or difference. Ore’s correspondence below is a separate fixed- U order isomorphism; it does not identify these semantic roles.

Ellerman uses the refinement order

$$\sigma \preceq \pi \iff \begin{array}{l} \text{each block of } \pi \text{ is contained} \\ \text{in some block of } \sigma. \end{array} \quad (4)$$

Finer partitions lie higher. The indiscrete partition $0_U = \{U\}$ is the bottom and the discrete partition

$$1_U = \{\{u\} : u \in U\} \quad (5)$$

is the top. Equivalently, refinement is inclusion of distinction sets:

$$\sigma \preceq \pi \iff \text{dit}(\sigma) \subseteq \text{dit}(\pi). \quad (6)$$

The join is the common refinement,

$$\pi \vee \sigma = \{B \cap C : B \in \pi, C \in \sigma, B \cap C \neq \emptyset\}, \quad (7)$$

and the meet is the greatest common coarsening. Its blocks can be obtained by taking overlap chains among blocks of π and σ , or equivalently by taking the reflexive-symmetric-transitive closure of $\text{indit}(\pi) \cup \text{indit}(\sigma)$, where $\text{indit}(\tau)$ is the indistinction set of pairs lying in the same block of τ .

This convention reverses the order commonly used for equivalence relations. With equivalence relations ordered by relation inclusion, the names “join” and “meet” in Eqs. (7) and the preceding description are exchanged. Ellerman’s convention is deliberate: it places the discrete partition at the top and makes refinement parallel to inclusion of ditsets.

C. Ore’s correspondence at the level of context generators

For finite U , the assignment

$$\pi \longleftrightarrow \mathcal{B}(\pi) \quad (8)$$

is a bijection between partitions of U and Boolean subalgebras of $\mathcal{P}(U)$. The finite restriction matters: for infinite U , partitions correspond not to all Boolean subalgebras of $\mathcal{P}(U)$, but to the complete atomic Boolean set algebras closed under arbitrary unions of their atoms. The blocks of π are the atoms of $\mathcal{B}(\pi)$. Semantically, however, the correspondence does not identify its two sides: π is one whole classification question, whereas $\mathcal{B}(\pi)$ is a family of membership-based event propositions. Write $\langle S \rangle_{\text{BA}}$ for the Boolean algebra generated by a family S . The following identities are the order-theoretic core of the comparison:

$$\sigma \preceq \pi \iff \mathcal{B}(\sigma) \subseteq \mathcal{B}(\pi), \quad (9)$$

$$\mathcal{B}(\pi \vee \sigma) = \langle \mathcal{B}(\pi) \cup \mathcal{B}(\sigma) \rangle_{\text{BA}}, \quad (10)$$

$$\mathcal{B}(\pi \wedge \sigma) = \mathcal{B}(\pi) \cap \mathcal{B}(\sigma). \quad (11)$$

Indeed, Eq. (9) makes the bijection an order isomorphism. Intersection is the greatest common Boolean subalgebra, and the Boolean algebra generated by a union is the least common Boolean superalgebra. This proves Eqs. (10) and (11). Equivalently, the atoms of the generated algebra in Eq. (10) are the nonempty intersections of blocks of π and σ , while the atoms of the intersection algebra in Eq. (11) are the blocks of their greatest common coarsening.

The two relational descriptions place closure on opposite operations. On the Boolean-subalgebra side, meet is literal intersection and join requires Boolean generation. On the distinction side,

$$\text{indit}(\pi \wedge \sigma) = \text{cl}_{\text{eq}}(\text{indit}(\pi) \cup \text{indit}(\sigma)), \quad (12)$$

$$\begin{aligned} \text{indit}(\pi \vee \sigma) &= \text{indit}(\pi) \cap \text{indit}(\sigma), \\ \text{dit}(\pi \vee \sigma) &= \text{dit}(\pi) \cup \text{dit}(\sigma), \end{aligned} \quad (13)$$

where cl_{eq} denotes reflexive-symmetric-transitive closure. Thus equivalence closure occurs at partition meet, while Boolean closure occurs at partition join.

There is also a canonical answer-to-question map that should not be confused with an order or lattice morphism:

$$\begin{aligned} q : \mathcal{P}(U) &\longrightarrow \Pi(U), \\ q(S) &= \begin{cases} \{S, U \setminus S\}, & \emptyset \subsetneq S \subsetneq U, \\ 0_U, & S = \emptyset \text{ or } S = U. \end{cases} \end{aligned} \quad (14)$$

It forgets which complementary answer was called yes: $q(S) = q(U \setminus S)$. For every partition π ,

$$S \in \mathcal{B}(\pi) \iff q(S) \preceq \pi. \quad (15)$$

Consequently, if

$$\mathcal{M} = \bigcup_k [0_U, \pi_k], \quad \mathcal{M}_{\leq 2} = \{\alpha \in \mathcal{M} : |\alpha| \leq 2\}, \quad (16)$$

then the concrete event carrier satisfies

$$L = q^{-1}(\mathcal{M}_{\leq 2}), \quad [0_U, \pi] \cap [0_U, \sigma] = [0_U, \pi \wedge \sigma]. \quad (17)$$

This is an exact bridge between event answers and binary partition questions, but it does not identify their ordered structures. If $\emptyset \subsetneq S \subsetneq T \subsetneq U$, then $S \subset T$ while the distinct binary partitions $q(S)$ and $q(T)$ are incomparable. Moreover, q identifies complementary events.

For a chosen family, Ore's correspondence acts on the context cover, not on its pasted carrier. It maps each π_k to $\mathcal{B}(\pi_k)$; in the concrete set representation one then takes the union of those event algebras, with equal subsets identified. Because that union is generally not itself a Boolean algebra, it is outside the image $\{\mathcal{B}(\tau) : \tau \in \Pi(U)\}$; Ore's correspondence therefore assigns no partition to L and does not turn pasting into a partition operation.

Equation (11) therefore shows that, for this fixed- U representation, the meet of the generating partitions encodes exactly the Boolean event algebra shared by two contexts. In particular,

$$\pi \wedge \sigma = 0_U \iff \mathcal{B}(\pi) \cap \mathcal{B}(\sigma) = \{\emptyset, U\}. \quad (18)$$

Thus $\pi \wedge \sigma \neq 0_U$ is equivalent to nontrivial overlap at the level of event propositions. This condition is weaker than atomic intertwining in the usual Greechie or Gleason sense. Atomic intertwining requires a block that is an outcome of both contexts,

$$\pi \cap \sigma \neq \emptyset, \quad (19)$$

where the intersection is taken between π and σ as sets of blocks; in Hilbert space the corresponding condition is a common atomic projection (a ray projection in a maximal rank-one context) [11].

Here and below, concatenation abbreviates a subset; for example, $12 = \{1, 2\}$. On $U = \{1, 2, 3, 4, 5, 6, 7, 8\}$, for example, let

$$\pi = \{12, 34, 56, 78\}, \quad \sigma = \{13, 24, 57, 68\}. \quad (20)$$

No block is shared, so $\pi \cap \sigma = \emptyset$, but

$$\begin{aligned} \pi \wedge \sigma &= \{1234, 5678\}, \\ \mathcal{B}(\pi) \cap \mathcal{B}(\sigma) &= \{\emptyset, 1234, 5678, U\}. \end{aligned} \quad (21)$$

The meet of the generating partitions therefore represents a common coarse-grained classification even though the contexts have no shared atomic outcome.

Equation (10) identifies the join with the Boolean closure of the two context algebras inside $\mathcal{P}(U)$. More generally,

$$\left\langle \bigcup_k \mathcal{B}(\pi_k) \right\rangle_{\text{BA}} = \mathcal{B} \left(\bigvee_k \pi_k \right). \quad (22)$$

The left-hand side is the Boolean algebra generated by L , not L itself; the join describes a closure of the concrete context cover rather than the pasted event structure.

That closure need not be an invariant of the abstract pasting. Consider three binary contexts given

by the coordinate partitions of bit strings. On $U_4 = \{000, 011, 101, 110\}$, the even-parity strings, their contextwise union is the horizontal sum of three four-element Boolean algebras, and their joint refinement has four atoms, so its Boolean closure has 16 elements. On $U_8 = \{0, 1\}^3$, the same three coordinate contexts again have an eight-element horizontal-sum carrier, isomorphic to the one on U_4 , but their joint refinement has eight atoms and the Boolean closure has 256 elements. Thus even isomorphic pasted carriers can have nonisomorphic ambient Boolean closures: the join summarizes the selected concrete realization, not the abstract pasted carrier alone.

For a concrete set-partition realization, the point valuations form an order-determining and separating family of two-valued states: each $u \in U$ defines

$$v_u(S) = \begin{cases} 1, & u \in S, \\ 0, & u \notin S. \end{cases} \quad (23)$$

If $S \not\subseteq T$, some $u \in S \setminus T$ satisfies $v_u(S) = 1 > 0 = v_u(T)$; thus these valuations determine the inherited order and separate distinct events. The concrete carrier's inclusion in the Boolean algebra of the joint classical refinement is an order embedding. When the pasted carrier is a lattice, this inclusion need not preserve its lattice operations; more generally, it need not preserve cross-context partial operations. This concrete subset representation underlies the mathematical equivalence with automaton and generalized-urn models.

There is a conditional converse. Let L be a finite orthoalgebra or orthomodular poset covered by finite Boolean contexts, and let $\mathfrak{S}$ be a family of globally defined two-valued states. Call $\mathfrak{S}$ *order determining* when

$$a \not\leq b \implies \text{some } v \in \mathfrak{S} \text{ has } v(a) = 1, v(b) = 0. \quad (24)$$

Then

$$\eta(a) = \{v \in \mathfrak{S} : v(a) = 1\} \quad (25)$$

is an order embedding into $\mathcal{P}(\mathfrak{S})$, and the images of the atoms in each context partition $\mathfrak{S}$. Thus the contexts acquire a faithful set-partition representation preserving their Boolean operations. Conversely, the point valuations of every faithful concrete partition representation are order determining [2, 3, 12]. Pairwise separation of abstract event labels is weaker than order determination; the triangle example below gives an explicit witness. Independently, an order-determining representing family need not exhaust all abstract two-valued states.

III. EXAMPLES AND THE LEVEL SHIFT

A. The four-state Chinese lantern

Let

$$U = \{1, 2, 3, 4\}, \quad \pi = \{12, 34\}, \quad \sigma = \{13, 24\}. \quad (26)$$

The induced event algebras are

$$\mathcal{B}(\pi) = \{\emptyset, 12, 34, U\}, \quad (27)$$

$$\mathcal{B}(\sigma) = \{\emptyset, 13, 24, U\}. \quad (28)$$

Their contextwise union is the six-element event structure traditionally called the Chinese-lantern logic. Its four middle elements belong to two alternative contexts and do not collectively form one partition. Cross-context Boolean operations must be distinguished from order operations. Although $12 \cap 13 = \{1\}$ in $\mathcal{P}(U)$, the event $\{1\}$ is absent from the carrier, so the contextual Boolean conjunction of 12 and 13 is undefined. In the underlying six-element inclusion poset, however, their order-theoretic meet exists and equals $\emptyset$.

At the level of whole partitions,

$$\pi \wedge \sigma = 0_U = \{1234\}, \quad (29)$$

$$\pi \vee \sigma = 1_U = \{1, 2, 3, 4\}, \quad (30)$$

where the last expression denotes four singleton blocks. These equations neither reproduce nor contradict the pasting; they summarize this concrete context pair. By Eqs. (18) and (10), the meet records its trivial overlap algebra, while the join records its full Boolean closure inside $\mathcal{P}(U)$, which distinguishes all four hidden states.

The example also separates two occurrences of non-distributivity. The set $\{0_U, \pi, \sigma, 1_U\}$ is a four-element Boolean algebra inside $\Pi(U)$, whereas the six-element pasted event structure is non-distributive. Conversely, the three $2 + 2$ partitions

$$\{12, 34\}, \quad \{13, 24\}, \quad \{14, 23\} \quad (31)$$

together with 0_U and 1_U form the five-element diamond lattice M_3 as a sublattice of $\Pi(U)$. The two nondistributivities occur at different logical levels.

B. The Firefly logic L_{12}

Consider

$$\pi = \{12, 34, 5\}, \quad \sigma = \{13, 24, 5\} \quad (32)$$

on $U = \{1, 2, 3, 4, 5\}$. Their event algebras are

$$\mathcal{B}(\pi) = \{\emptyset, 12, 34, 5, 1234, 125, 345, U\}, \quad (33)$$

$$\mathcal{B}(\sigma) = \{\emptyset, 13, 24, 5, 1234, 135, 245, U\}. \quad (34)$$

Their common part is

$$\mathcal{B}(\pi) \cap \mathcal{B}(\sigma) = \{\emptyset, 5, 1234, U\}. \quad (35)$$

Consequently their contextwise union has $8 + 8 - 4 = 12$ elements and is the pasted event structure known as the Firefly logic L_{12} .

Ellerman's operations give

$$\pi \wedge \sigma = \{1234, 5\}, \quad (36)$$

$$\pi \vee \sigma = 1_U. \quad (37)$$

The meet is the partition representing the shared binary question “5 or not 5,” since

$$\mathcal{B}(\pi \wedge \sigma) = \mathcal{B}(\pi) \cap \mathcal{B}(\sigma). \quad (38)$$

Here the overlap is also atomic because $\pi \cap \sigma = \{5\}$. The meet records the Boolean algebra generated by the shared outcome and its complement, while the join records the complete hidden-state refinement. The twelve event propositions do not become twelve partition elements; their generating contexts are described one level higher.

C. A three-context stress test: triangular incidence

A cyclic cover with nontrivial pairwise overlaps is obtained on $U = \{1, 2, 3, 4\}$ from

$$\begin{aligned} \pi_1 &= \{1, 24, 3\}, & \pi_2 &= \{3, 14, 2\}, \\ \pi_3 &= \{2, 34, 1\}. \end{aligned} \quad (39)$$

Their context incidence is triangular: π_1 and π_2 share the outcome block 3, π_2 and π_3 share 2, and π_3 and π_1 share 1. Ellerman’s pairwise meets are

$$\begin{aligned} \pi_1 \wedge \pi_2 &= \{3, 124\}, \\ \pi_2 \wedge \pi_3 &= \{2, 134\}, \\ \pi_3 \wedge \pi_1 &= \{1, 234\}, \end{aligned} \quad (40)$$

whereas

$$\pi_i \vee \pi_j = 1_U \quad (i \neq j), \quad \bigwedge_{i=1}^3 \pi_i = 0_U. \quad (41)$$

By Eq. (11), the pairwise context intersections are the four-element Boolean algebras generated by the three binary partitions in Eq. (40), while the triple intersection is $\{\emptyset, U\}$. Writing $\mathcal{B}_i = \mathcal{B}(\pi_i)$, inclusion–exclusion gives the concrete set-represented carrier

$$\begin{aligned} L_{\Delta}^{\text{conc}} &= \mathcal{B}_1 \cup \mathcal{B}_2 \cup \mathcal{B}_3 \\ &= \mathcal{P}(U) \setminus \{\{4\}, \{1, 2, 3\}\}, \\ |L_{\Delta}^{\text{conc}}| &= 14. \end{aligned} \quad (42)$$

This concrete ordered carrier is not the abstract Boolean-block amalgam determined only by those context overlaps. Inherited inclusion adds, for example,

$$\{1\} \subset \{1, 4\}, \quad \{2\} \subset \{2, 4\}, \quad \{3\} \subset \{3, 4\}, \quad (43)$$

although, in each displayed inclusion, the two events belong to different contexts. Thus 14, 24, and 34 are context atoms but not global atoms of the concrete inclusion order. With $x^\perp = U \setminus x$, that order is not orthomodular: for $x = 1 \subset y = 14$, the event 4 is absent, so $y \wedge x^\perp = 14 \wedge 234 = \emptyset$ and $x \vee (y \wedge x^\perp) = x \neq y$.

The three one-outcome-per-context equations have exactly four solutions, namely the point valuations v_u of

Eq. (23); they therefore exhaust the two-valued states of the abstract amalgam. Every local outcome is selected by at least one of them, and together they distinguish all 14 event labels. Nevertheless, they are not order determining for the abstract order: the events 1 and 14 are incomparable there, yet

$$\eta(1) = \{v_1\} \subset \{v_1, v_4\} = \eta(14), \quad (44)$$

Thus the state-set representation produces precisely the extra inclusion in Eq. (43). For the resulting inherited subset order, the same point valuations are order determining. Cyclic atomic intertwining and nontrivial pairwise meets do not by themselves produce state scarcity.

IV. ADDITIONAL OPERATIONS IN ELLERMAN’S FRAMEWORK

A. Implication

Ellerman’s implication has no counterpart as an operation of contextual pasting: that construction defines no operation taking two context partitions to a partition. Ellerman defines $\sigma \Rightarrow \pi$ by examining every block B of π : if B is contained in a block of σ , B is discretized; otherwise it remains whole. The operation satisfies

$$\sigma \Rightarrow \pi = 1_U \iff \sigma \preceq \pi. \quad (45)$$

If the partitions arise from random variables or experiments, the discrete parts of $\sigma \Rightarrow \pi$ mark the π -outcomes that already determine the σ -outcome. This expresses deterministic functional dependence of the σ -outcome on the π -outcome and is close in spirit to automaton-style state identification. In a fixed hidden-state representation, it may provide a way to organize implication chains across the intertwined contexts of Hardy-type arguments; establishing such an application requires additional structure beyond Ore’s correspondence [13].

B. Boolean-indexed operations and their limitation

Ellerman’s graph method associates a partition operation with any Boolean truth function. Edges of the complete graph on U are labeled true or false according to whether their endpoints are distinguished by each input partition; edges on which the truth function evaluates false are retained, and connected components determine the output partition [7]. Equivalently, a relation built from dits and indits is closed reflexively, symmetrically, and transitively.

The construction differs from a compositional copy of Boolean logic. The closure step means that composing the resulting partition operations need not reproduce composition of their indexing truth functions. Ellerman notes, for example, that a compound operation need not reduce to one of the sixteen basic binary operations, and

Boolean tautologies such as Peirce's law can fail as partition tautologies.

Ellerman's implication also supplies negations. For every fixed $\pi \in \Pi(U)$, the relative π -negation

$$\neg_\pi \sigma := \sigma \Rightarrow \pi, \quad (46)$$

is defined for every $\sigma \in \Pi(U)$. The π -regular elements $x = \sigma \Rightarrow \pi$ form a Boolean core in $[\pi, 1_U]$, with bottom π and top 1_U . Within that core the complement of x is

$$\neg_\pi x = x \Rightarrow \pi = (\sigma \Rightarrow \pi) \Rightarrow \pi. \quad (47)$$

The special choice $\pi = 0_U$ gives Ellerman's globally defined partition negation

$$\neg \sigma := \sigma \Rightarrow 0_U = \begin{cases} 1_U, & \sigma = 0_U, \\ 0_U, & \sigma \neq 0_U. \end{cases} \quad (48)$$

Its Boolean core is the two-element algebra $\{0_U, 1_U\}$. It is therefore a Boolean complement on that core, but not a lattice complement for an intermediate partition σ , since $\sigma \vee \neg \sigma = \sigma \neq 1_U$. This distinguishes a negation defined on all of $\Pi(U)$ from Boolean complementation on the whole partition lattice [5, 6]. The graph method is a truth-table-indexed definition scheme whose closure operation creates non-Boolean behavior.

V. DIRECT-SUM DECOMPOSITIONS AND CONTEXTUALITY

A. The partial partition algebra of DSDs

Throughout this section V is finite dimensional. Ellerman's vector-space analogue of a partition is a DSD

$$\pi = \{V_i\}_{i \in I}, \quad V = \bigoplus_{i \in I} V_i. \quad (49)$$

When it arises from a diagonalizable operator, the DSD records the eigenspace classification while abstracting from the numerical eigenvalues. The set $\text{DSD}(V)$ is a meet-semilattice under refinement [8]. Its bottom element is the indiscrete DSD $0_V = \{V\}$. A maximal DSD

$$\omega = \{L_1, \dots, L_n\} \quad (50)$$

consists of one-dimensional subspaces and determines a local partition algebra

$$\mathcal{I}(\omega) := [0_V, \omega] \subseteq \text{DSD}(V), \quad \mathcal{I}(\omega) \cong \Pi(\{1, \dots, n\}). \quad (51)$$

The isomorphism groups the component lines according to a partition of their index set. Every DSD can be refined by choosing bases in its components, so

$$\text{DSD}(V) = \bigcup_{\substack{\omega \in \text{DSD}(V) \\ \omega \text{ maximal}}} \mathcal{I}(\omega). \quad (52)$$

Two DSDs $\{V_i\}$ and $\{W_j\}$ of the same ambient vector space V are compatible if their nonzero intersections

$$\{V_i \cap W_j : V_i \cap W_j \neq \{0\}\} \quad (53)$$

span V . Only then is their join defined. The absence of a common upper bound is the DSD manifestation of incompatibility.

For a selected DSD family $\{\alpha_k\}$, the interval union

$$\mathcal{M}_{\text{DSD}} := \bigcup_k [0_V, \alpha_k] \quad (54)$$

has the same form as the selected set-side cover $\mathcal{M}$ in Eq. (16). Under literal linearization, $[0_U, \pi_k] \cong [0_V, \hat{\pi}_k]$. This interval analogy concerns the local-to-global architecture; it does not identify the local event carriers. The intervals are generally non-Boolean and overlap through common coarsenings, whereas projection-event contexts overlap through common propositions. The full cover in Eq. (52) is broader: literal linearizations of partitions of one U all lie in the single coordinate interval $\mathcal{I}(\omega_U)$, while $\text{DSD}(V)$ contains incompatible maximal DSDs.

Harding's related decomposition construction retains ordered complementary factor pairs, with orthocomplementation given by interchanging the two factors [14]. An unordered binary DSD instead packages complementary answers into one decomposition. In the Hilbert-space specialization, orthogonality selects M and $M^\perp$ and yields self-adjoint projection events.

B. Orthogonal vector-space partitions and exclusivity

A finite-dimensional projective measurement, equivalently the spectral family of a self-adjoint observable, has a precise orthogonal-DSD description. It is a maximal context when all its nonzero spectral projections have rank one; a degenerate spectral family is a coarser projective measurement. For a self-adjoint observable with spectral resolution

$$A = \sum_i a_i P_i, \quad (55)$$

put $V_i = \text{ran}(P_i)$, the range of P_i . Its spectral family satisfies

$$\mathcal{H} = \bigoplus_i V_i, \quad P_i P_j = 0 \ (i \neq j), \quad \sum_i P_i = I. \quad (56)$$

Thus $\{V_i\}$ is an orthogonal DSD and $\{P_i\}$ is a partition of the identity. Pairwise orthogonality encodes exclusivity of distinct outcomes, while the resolution of the identity encodes completeness. More generally, an orthogonal DSD $\alpha = \{V_i\}_{i \in I}$ determines the Boolean algebra

$$\mathcal{C}_\alpha = \left\{ \sum_{i \in S} P_i : S \subseteq I \right\} \quad (57)$$

of projection events. For finite orthogonal DSDs α and β ,

$$\alpha \preceq \beta \iff \mathcal{C}_\alpha \subseteq \mathcal{C}_\beta, \quad \mathcal{C}_{\alpha \wedge \beta} = \mathcal{C}_\alpha \cap \mathcal{C}_\beta. \quad (58)$$

Indeed, a nontrivial projection P belongs to $\mathcal{C}_\alpha$ precisely when the binary DSD $\{\text{ran}(P), \text{ran}(I - P)\}$ is a coarsening of α ; the endpoint projections correspond to 0_V . The second identity then follows from the universal property of the meet. Projection Boolean algebras may be pasted along all common projections, including sums of component projections. This is the usual projection-event construction and is distinct from the partial partition algebra $\text{DSD}(V)$ [8, 11].

It is worth stressing that an orthogonal DSD is not obtained by taking equivalence classes under vector orthogonality. Orthogonality is not reflexive on nonzero vectors, and it is not transitive. In $\mathbb{C}^3$, for example,

$$e_1 \perp e_2, \quad e_2 \perp (e_1 + e_3), \quad e_1 \not\perp (e_1 + e_3). \quad (59)$$

Nor do the subspaces in Eq. (56) partition the underlying set of vectors. A generic vector has a unique component decomposition $v = \sum_i v_i$ with $v_i \in V_i$, but it need not belong to any single V_i . Unique decomposition replaces the unique block membership of a set partition.

After choosing an orthonormal basis adapted to one DSD, its basis rays can be grouped into a genuine set partition whose block spans are the V_i . Incompatible DSDs, however, need not possess a common adapted basis. If an entire family is refined by one coordinate DSD, all of its members are simultaneously diagonalizable and the construction reduces to the classical case described below.

Orthogonality can nevertheless serve as the primitive relation through biorthogonal closure rather than equivalence closure. For $S \subseteq \mathcal{H}$, define

$$S^\perp = \{x \in \mathcal{H} : \langle x, s \rangle = 0 \text{ for every } s \in S\}. \quad (60)$$

The fixed points of $S \mapsto S^{\perp\perp}$ are the closed linear subspaces that form the usual projection logic. Equivalently, one may use an orthogonality hypergraph whose vertices are rays (or, more generally, projections) and whose contexts are maximal orthogonal families resolving the identity. The resulting statement is therefore that quantum logic is generated by orthogonal decompositions, not that orthogonality partitions all vectors into equivalence classes.

C. Why set-partition models remain classical

The literal linearization of a finite set U takes a vector space with basis $\{e_u : u \in U\}$ and maps

$$\pi = \{B_i\} \longmapsto \hat{\pi} = \{\text{span}\{e_u : u \in B_i\}\}_i. \quad (61)$$

Every literal linearization of a partition of this fixed U is refined by the same maximal coordinate DSD

$$\omega_U = \{\text{span}(e_u) : u \in U\}. \quad (62)$$

Hence all literal linearizations of partitions on the same finite set U are mutually compatible; their DSD joins are the literal linearizations of the corresponding joint set refinements $\bigvee_k \pi_k$. Equivalently, all lie below one maximal coordinate DSD. This reflects the fact that automaton and urn partition logics are concrete classical models with separating two-valued states.

For example, literal linearization of Eq. (32) gives

$$\hat{\pi} = \{\text{span}(e_1, e_2), \text{span}(e_3, e_4), \text{span}(e_5)\}, \quad (63)$$

$$\hat{\sigma} = \{\text{span}(e_1, e_3), \text{span}(e_2, e_4), \text{span}(e_5)\}. \quad (64)$$

Their join is ω_U , so they can be represented by commuting diagonal operators. The operational Firefly event structure does not include that singleton refinement as an available joint experiment.

The triangle family in Eq. (39) makes this compatibility issue especially explicit. Put $L_j = \text{span}(e_j)$ and $V_{ij} = L_i \oplus L_j$. Its literal linearizations in $\mathbb{C}^4$ are

$$\begin{aligned} \hat{\pi}_1 &= \{L_1, V_{24}, L_3\}, \\ \hat{\pi}_2 &= \{L_3, V_{14}, L_2\}, \\ \hat{\pi}_3 &= \{L_2, V_{34}, L_1\}. \end{aligned} \quad (65)$$

These are complete orthogonal three-outcome DSDs of rank pattern $(1, 2, 1)$, not maximal rank-one contexts. All are coarsenings of the same coordinate DSD $\omega_U = \{L_1, L_2, L_3, L_4\}$, every pair has join ω_U , and their component projections are diagonal and mutually commuting. Thus cyclically shared outcomes do not by themselves yield quantum incompatibility.

Nor does the corresponding six-vertex incidence hypergraph have a faithful realization by three distinct maximal orthogonal triples in $\mathbb{C}^3$. The three shared rays occur pairwise in contexts and therefore form an orthogonal basis; the private ray in each triple is then forced to coincide with the remaining shared ray. Ellerman's unrestricted nonorthogonal DSDs can realize the incidence by distinct bases, but those bases are not quantum contexts. A maximal orthogonal realization in higher dimension requires refining the private coarse outcomes and therefore represents a different hypergraph.

D. Intertwining contexts in the Gleason sense

In the Gleason and Kochen–Specker setting, a context is a maximal Boolean algebra of commuting projections; equivalently, its atoms form a maximal pairwise-orthogonal family resolving the identity [15, 16]. If two orthonormal contexts share a ray L ,

$$\omega = \{L, L_2, \dots, L_n\}, \quad (66)$$

$$\omega' = \{L, L'_2, \dots, L'_n\}, \quad (67)$$

then the binary DSD

$$\alpha_L = \{L, L^\perp\} \quad (68)$$

is refined by both contexts and therefore belongs to $\mathcal{I}(\omega) \cap \mathcal{I}(\omega')$. In this sense, Ellerman's DSD structure can encode an observable occurring in several contexts.

The common event need not be atomic. In $\mathbb{C}^4$, write $L_i = \text{span}(e_i)$ and $K_{ij}^\pm = \text{span}((e_i \pm e_j)/\sqrt{2})$. Put

$$\begin{aligned}\gamma &= \{L_1, L_2, L_3, L_4\}, \\ \gamma' &= \{K_{12}^+, K_{12}^-, K_{34}^+, K_{34}^-\}.\end{aligned}\quad (69)$$

The contexts share no ray. With $M_{ij} = \text{span}(e_i, e_j)$ and P_M denoting the orthogonal projection onto M ,

$$\gamma \wedge \gamma' = \{M_{12}, M_{34}\}, \quad \mathcal{C}_\gamma \cap \mathcal{C}_{\gamma'} = \{0, P_{M_{12}}, P_{M_{34}}, I\}. \quad (70)$$

Thus DSD meets detect shared coarse projections as well as shared rays; atomic intertwining is the stronger condition.

A Hilbert-space incidence model of L_{12} can therefore be made in $\mathcal{H} = \mathbb{C}^3$. Let $L_5 = \text{span}(e_3)$ and choose

$$\omega = \{\text{span}(e_1), \text{span}(e_2), L_5\}, \quad (71)$$

$$\omega' = \left\{ \text{span}\left(\frac{e_1 + e_2}{\sqrt{2}}\right), \text{span}\left(\frac{e_1 - e_2}{\sqrt{2}}\right), L_5 \right\}. \quad (72)$$

These maximal orthogonal DSDs satisfy

$$\omega \wedge \omega' = \{L_5, L_5^\perp\}, \quad \omega \vee \omega' \text{ is undefined.} \quad (73)$$

Here the protojoin contains only the shared ray L_5 and therefore does not span $\mathcal{H}$. More generally, two maximal one-dimensional DSDs have a join only when they are equal: their nonzero line intersections span $\mathcal{H}$ only when they share a complete basis. Hence in the Firefly example the nontrivial meet carries the shared incidence information, while the undefined join records that the maximal contexts are distinct and incompatible. They thus reproduce the incidence pattern of two incompatible contexts with a shared outcome. This is not the literal linearization of the five hidden states; it retains only the contextual incidence pattern. Constructing such graph-incidence realizations by faithful orthogonal representations is studied systematically in Ref. [11].

E. Additional structure for quantum-foundational applications

Ellerman defines DSDs for arbitrary direct sums, not necessarily the orthogonal partitions of the identity in Eq. (56). His partial order and partition operations do not by themselves encode the orthogonality relation on rays that carries Gleason and Kochen–Specker arguments. To address those problems one must at least restrict to orthogonal DSDs and retain their component projections.

There is also a level mismatch. The DSD $\{L, L^\perp\}$ packages the two complementary events P_L and $I - P_L$ into one whole measurement. In a projection logic they

are separate propositions. A Gleason measure requires a context-independent assignment

$$\mu(P) \geq 0, \quad \sum_{L_i \in \omega} \mu(P_{L_i}) = 1, \quad (74)$$

with the same $\mu(P)$ whenever P occurs in several contexts. For a finite-dimensional real or complex Hilbert space of dimension at least three, every normalized measure on all projections that is finitely additive on mutually orthogonal families has the form

$$\mu(P) = \text{Tr}(\rho P) \quad (75)$$

for a unique $\rho \geq 0$ with $\text{Tr} \rho = 1$, where Tr denotes the trace. In finite dimension, countable additivity adds no further requirement. This probability assignment lives on the component projections, not on whole DSDs alone.

Indeed, each orthogonal maximal DSD ω determines the ordinary Boolean projection algebra $\mathcal{C}_\omega$ of Eq. (57). For the two contexts in Eq. (72),

$$\mathcal{C}_\omega \cap \mathcal{C}_{\omega'} = \{0, P_5, I - P_5, I\}, \quad (76)$$

so their pasted union has $8 + 8 - 4 = 12$ elements and is isomorphic to L_{12} . This recovery uses the usual event structure obtained by pasting projection contexts. The DSD theory organizes the measurements one level higher; it does not replace the orthogonality and probability structures needed for quantum foundations.

F. Scarcity of two-valued states

A separate issue concerns states rather than partition operations. Ellerman's native carrier consists of whole DSDs, whereas a Kochen–Specker two-valued state assigns values to their component projections. A homomorphism taking whole partitions to 0 or 1 would be a different semantics from a dispersion-free assignment of exactly one outcome in each measurement context.

The required state layer can be adjoined as follows. Let Ω be a finite family of maximal orthogonal DSDs and let $A(\Omega) = \bigcup_{\omega \in \Omega} \omega$ be the set of their component rays, with a shared ray represented by the same element in every context. An atomic two-valued assignment is a map satisfying

$$v : A(\Omega) \longrightarrow \{0, 1\}, \quad \sum_{L \in \omega} v(L) = 1 \quad (\omega \in \Omega). \quad (77)$$

Within each context it extends uniquely by finite additivity:

$$\tilde{v}_\omega \left(\sum_{L \in S} P_L \right) = \sum_{L \in S} v(L), \quad S \subseteq \omega. \quad (78)$$

It defines a state on the pasted projection-event carrier only when these contextwise extensions agree on every

projection in $\mathcal{C}_\omega \cap \mathcal{C}_{\omega'}$. Equation (77) alone ensures this agreement when every overlap is generated by shared rays, as in the Firefly example; otherwise agreement on shared non-atomic projections is an additional global constraint. For Eqs. (69) and (70), for example, selecting $\text{span}(e_1)$ in γ and a ray in M_{34} in γ' satisfies both local one-ray equations but assigns respectively 1 and 0 to $P_{M_{12}}$. Write $\mathcal{V}(\Omega)$ for the set of context-consistent assignments. At the atomic level, this family is *unital* if every $L \in A(\Omega)$ receives value 1 in some $v \in \mathcal{V}(\Omega)$, and *separating* if every pair of distinct rays is distinguished by some v . Failure of the first property is nonunitality; failure of the second is nonseparability. At the limiting Kochen–Specker case,

$$\mathcal{V}(\Omega) = \emptyset, \quad (79)$$

because no context-independent 0–1 assignment obeys all of the one-outcome-per-context constraints [16, 17].

The Firefly logic provides a control example. For $\omega = \{L_1, L_2, L_5\}$ and $\omega' = \{L_3, L_4, L_5\}$, put $\Omega_{12} = \{\omega, \omega'\}$. Equation (77) has

$$|\mathcal{V}(\Omega_{12})| = 1 + 2 \cdot 2 = 5. \quad (80)$$

One state selects the shared ray L_5 ; the other four set $v(L_5) = 0$ and select one ray from each of the remaining pairs. Hence L_{12} is both unital and separating despite its intertwined contexts. Intertwining and incompatibility therefore do not by themselves imply state scarcity.

For a literal set-partition model, the point valuations in Eq. (23) are automatically unital and separate all events. Consequently, a nonunital, nonseparating, or Kochen–Specker event structure cannot have a faithful concrete representation of that kind. Nor can scarcity be inferred merely from pairwise DSD meets or undefined joins: it is a global consistency property of the entire orthogonality hypergraph. The general DSD(V) framework represents this hierarchy only after restriction to orthogonal DSDs and imposition of the global constraint Eq. (77) together with the overlap-consistency condition following Eq. (78). The DSDs in this orthogonal specialization still organize complete measurement contexts one level higher, while state scarcity is a global property of assignments on the pasted projection event structure.

VI. STRUCTURAL COMPARISON AND PROSPECTS

The fixed- U comparison has three exact components. Ore’s correspondence maps each whole-partition generator π_k to its Boolean event context $\mathcal{B}(\pi_k)$; the map q in Eq. (14) sends an event proposition to its binary partition question; and the interval cover $\mathcal{M}$ retains all coarsenings of the selected questions. None of these constructions turns the union of event contexts into an operation on $\Pi(U)$. The linear analogue has the same interval architecture, while Eq. (58) states that the projection algebra

of the greatest common orthogonal-DSD coarsening is exactly the intersection of the two projection-event algebras.

Other whole-partition quantities require a separate interpretation. For the uniform distribution on the chosen finite U , Ellerman’s logical entropy and logical mutual information are, respectively,

$$h(\pi) = \frac{|\text{dit}(\pi)|}{|U|^2}, \quad m(\pi, \sigma) = \frac{|\text{dit}(\pi) \cap \text{dit}(\sigma)|}{|U|^2}. \quad (81)$$

In general $m(\pi, \sigma) \neq h(\pi \wedge \sigma)$, because equivalence closure in Eq. (12) gives

$$\text{dit}(\pi \wedge \sigma) \subseteq \text{dit}(\pi) \cap \text{dit}(\sigma), \quad (82)$$

possibly strictly. For the Chinese-lantern partitions,

$$\text{dit}(\pi) \cap \text{dit}(\sigma) = \{(1, 4), (4, 1), (2, 3), (3, 2)\}, \quad (83)$$

so $m(\pi, \sigma) = 1/4$, whereas $h(\pi \wedge \sigma) = h(0_U) = 0$ and the shared event algebra is only $\{\emptyset, U\}$. Nonzero overlap of distinctions can therefore coexist with no nontrivial common event proposition [6]. Whether partition implication can organize Hardy-type forcing chains, or whether these entropy quantities admit a useful operational reading for a pasted model, remains a representation-dependent research question rather than a consequence of Ore’s correspondence.

VII. CONCLUSION

The distinction is between the logic *of* partitions and contextual event structures *generated by* families of partitions, conventionally called partition logics. In the former, whole partitions are classifications or abstract attributes whose elementary logical instances concern same/different distinctions between pairs; the partitions admit refinement, join, meet, implication, and other closure-based operations. In the latter, partitions label experimental questions or contexts and their block unions, regarded as subsets of the hidden-state set, support membership-event propositions in a pasted event structure.

Ore’s correspondence is exact at the level of context generators but does not identify the two theories. It maps each π_k to $\mathcal{B}(\pi_k)$, while the canonical map q sends each event answer to its unordered binary partition question and identifies complementary answers. Since L is generally not a Boolean algebra $\mathcal{B}(\tau)$, neither construction turns pasting into a partition connective. Within a fixed- U representation, $\mathcal{B}(\pi \wedge \sigma) = \mathcal{B}(\pi) \cap \mathcal{B}(\sigma)$ gives the complete overlap algebra of a selected context pair, while $\mathcal{B}(\pi \vee \sigma) = \langle \mathcal{B}(\pi) \cup \mathcal{B}(\sigma) \rangle_{\text{BA}}$ gives its Boolean closure inside $\mathcal{P}(U)$. For the four-state Chinese-lantern realization, meet 0_U records a trivial context intersection and join 1_U records the Boolean closure that separates all four hidden states. More generally, the bit-string parity comparison shows that the closure can depend on the

chosen concrete realization. The triangle-shaped family exhibits a different limitation: its three pairwise meets recover the cyclic shared outcomes, but its concrete subset order is not the abstract block amalgam and its literal orthogonal DSDs remain compatible. More generally, a faithful set-partition representation requires an order-determining family of two-valued states; mere separation of event labels need not suffice. For the Firefly event structure L_{12} , the meet $\{1234, 5\}$ records the common binary classification generated by the shared atom and its complement. Atomic intertwining is an additional incidence condition on the original contexts, expressed by a common block or, in Hilbert space, a common ray.

An orthogonal DSD supplies the corresponding vector-based notion of a partition: its component projections form a partition of the identity, orthogonality gives exclusivity, and their sum to I gives completeness. It does not partition the set of vectors into equivalence classes; its defining property is the unique direct-sum decomposition of each vector.

Ellerman's DSD construction generalizes this organization to whole decompositions or abstract observables and allows incompatible DSDs to lack a join. For applications to Gleason or Kochen–Specker contextuality, the general DSD(V) framework must be restricted to orthogonal DSDs, passed to the associated Boolean algebras of component projections, and supplied with global state constraints. With that additional structure, the Firefly

incidence pattern is recovered as an event structure obtained by pasting projection contexts, and state scarcity up to the Kochen–Specker case can be expressed. In a concrete fixed- U realization, the same selected partitions can be viewed as whole-partition objects or as context generators; their carriers, operations, and physical uses nevertheless remain distinct.

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